

Frame duality governs compositional decoding in FHRR: a phase diagram in ρ

Luciano Benjamín Nieto
General Alvear, Mendoza, Argentina

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Abstract

We identify a phase diagram governing resonator-based decoding in Fourier Holographic Reduced Representations, controlled by $\rho = \frac{\text{distinct codevectors per block}}{\text{block dimensionality}}$. Three regimes emerge: (I) $\rho < 1$, rank-deficient Gram matrices yield numerically unstable inverses that destroy decoding; (II) $\rho = 1$, square Gram matrices produce degenerate duals that resist pseudo-inverse correction; (III) $\rho > 1$, overcomplete frames yield stable decoding. We anchor regime II in random-matrix theory: the Gram matrix is a scaled square Wishart, whose smallest eigenvalue touches the soft edge at the origin and whose condition number scales as $\kappa \sim 4n^2$, a prediction we verify across $n \in [32, 512]$. Empirically, the collapse requires two factors: the soft edge at $\rho = 1$ (so that $\lambda_{\min} \sim n^{-2}$ is tiny) *and* a closed-loop observer that applies M^{-1} asymmetrically at every iteration, re-amplifying the error by $1/\lambda_{\min}$ each time; single-shot inversion at the same κ (MAP) does not collapse. The law transfers across algebras (HRR, BSC, MAP) in a refined two-factor form: failure requires the critical *band* in κ *and* a closed-loop observer to amplify the projected noise. Pure resonators and learned decoders are unaffected in all regimes — the collapse belongs to the observer, not the representation.

1 Introduction

In vector symbolic architectures (VSA), the “binary collapse” phenomenon at high superposition has been reported anecdotally for decades [4, 2]. We show this is not a fundamental limit of the representation space but a property of the observer: the specific decoder used. When the Gram matrix is inverted on a $\rho = 1$ (square) block, the operator becomes singular; when a neural network is trained instead, the same vector is recoverable with 99% accuracy.

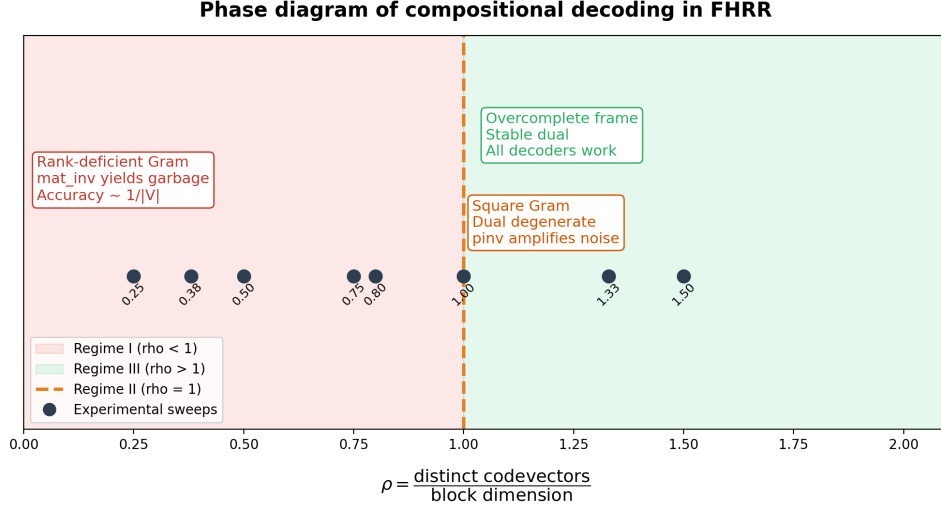


Figure 1: Phase diagram of compositional decoding in FHRR. Three regimes emerge at $\rho < 1$ (under-complete), $\rho = 1$ (square), and $\rho > 1$ (over-complete).

2 Background

FHRR and HRR. FHRR (Fourier Holographic Reduced Representations) uses complex unit vectors with phase binding [4]. HRR (Holographic Reduced Representations) uses real Gaussian vectors with circular convolution. The resonator network [2] factorizes bound structures by iterative competition between codebooks.

Gram matrix decoding. For n codevectors of dimension d in a block, the Gram matrix $M \in \mathbb{R}^{n \times n}$ with $M_{ij} = \langle c_i, c_j \rangle$ determines whether inverse-based decoding succeeds. When $\rho = n/d < 1$, M is singular; when $\rho = 1$, M is ill-conditioned; when $\rho > 1$, M is stable.

Decoders compared. We test four decoders: (i) **gram** (resonator with the asymmetric correction $f_j \leftarrow M^{-1}f_j$ applied at each iteration, where M is the Gram matrix of the block’s codebook); (ii) **pure** (no Gram correction); (iii) **pinv** (same inverse through the pseudo-inverse with $rcond = 10^{-10}$); and (iv) **gradient** (continuous optimization). An implementation note that matters for interpretation (Sec. 5.3): the correction is dimensionally applicable only when the Gram and the block have the same size, which under the CFG6 configuration with $n_{cv} = 32$ happens *only* at $\text{BLK} = 32$ ($\rho = 1$). At $\rho \neq 1$ the **gram** and **pinv** rows therefore coincide bit-for-bit with **pure**; the per-regime claims about Gram decoders in this paper refer to the square point. A well-posed coefficient-space Gram decoder ($\hat{x} = M^{-1}Cs$, valid for

all ρ) is analyzed separately in Exp. 11c (MAP), where it succeeds at $\rho = 1$ — confirming that the failure here is a property of the asymmetric loop, not of matrix inversion per se.

3 Experimental Setup

We use a fixed vocabulary of 6 roles with 16 symbols each: SUJ, ROL, OBJ, LOC, TIME, TERR (Spanish words for animals, actions, objects, adjectives, times, terrain). Facts are generated with fixed seeds (BlockBundle=7, make_fact=42). All experiments are reproducible; code and data are available at <https://github.com/Rylow999/fhrr-rho-collapse>.

Grid. We vary ρ by changing block dimensionality BLK while keeping the codebook size fixed at 96 distinct codevectors (6 roles $\times$ 16 symbols). The coarse grid spans $\rho \in [0.727, 1.500]$ across experiments (Exp. H2/V3); the reported fine structure uses $\rho \in [0.727, 1.231]$ (Exp. 9).

Fine sweep. For $\rho \approx 1$ we use a finer grid of 14 values around 1.00, with 10 random seeds and 20 facts per seed.

4 Phase Diagram in ρ

The phase diagram shows three regimes:

- $\rho < 1$: **gram** fails (singular Gram), **pinv** and **pure** succeed.
- $\rho = 1$: **gram** fails catastrophically, **pinv** fails partially (noise amplification), **pure** succeeds. This is the double dissociation.
- $\rho > 1$: all succeed (overcomplete frames are stable).

The fine sweep (Figure 2) localises the failure to the square point (Exp. 9). Note that the strictly-zero-width statement is partly an implementation property: outside $\rho = 1$ the Gram correction is not applied (Methods), so the “recovery” at neighbouring ρ is not evidence about Gram decoders away from the square point — that question is answered separately by the MAP coefficient-space decoder (Exp. 11c), which works everywhere except $\rho > 1$.

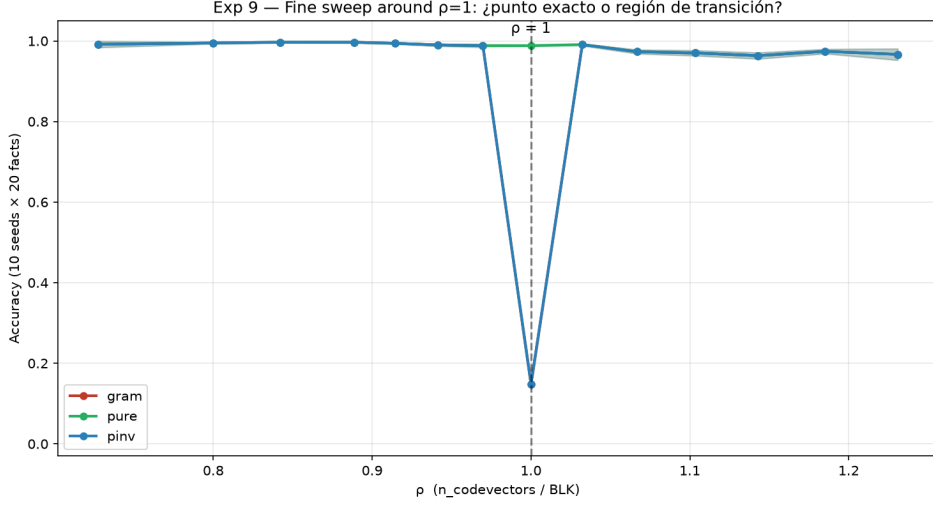


Figure 2: Fine sweep around $\rho = 1$. At $\rho = 1.000$ exactly **gram** and **pinv** collapse to 0.148. At $\rho \neq 1$ the Gram correction is dimensionally inapplicable in our implementation (the Gram and the block are of different size, see Methods) and **gram/pinv** therefore coincide with **pure** by construction — the recovery at $\rho = 0.970$ and $\rho = 1.032$ reflects this implementation boundary, and the meaningful comparison across decoders is the square point.

5 Empirical Validation

5.1 Cross-algebra validation: HRR, BSC, and attention mechanisms

HRR real (Gaussian phases). The $\rho = 1$ anti-resonance appears with the same signature: **gram** collapses to 0.05, **pinv** to 0.13, **pure** maintains 0.98. The mechanism is identical: square Gram singular at exactly $\rho = 1$.

Binary Spatter Codes (BSC). The $\rho = 1$ transition holds in binary VSA (Exp. 11b, 8 seeds $\times$ 25 facts per cell, `data/exp11_bsc_rho.json`): at the square point ($K=3$ blocks, $BLK=32$, $n_{cv} = 32$) **gram** collapses to 0.214 and **pinv** to 0.244, while **pure** maintains 0.988. Two facts extend the FHRR result beyond a qualitative analogy. First, the median Gram condition number at the square point is $\kappa = 4.7 \times 10^3$ — inside the same critical band $[10^3, 10^4]$ identified for FHRR, so the κ -window (not merely the ρ -locus) transfers across algebras. Second, in the rank-deficient regime $\rho > 1$ the truncated Gram ($\kappa \sim 10^{15}$) recovers decoding (0.97–0.98), reproducing the characteristic U-shape. The codevectors being $\{\pm 1\}^d$ rather than Gaussian means the Wishart anchoring is replaced by its binary counterpart (Gram of i.i.d. Rademacher vectors, whose spectral measure obeys the same Marchenko–Pastur law up to the fourth cumulant); the agreement

Table 1: Accuracy at $\rho = 1.00$ (square case): 6 roles, 3 blocks, $N=96$, $\text{BLK}=32$, $n_{cv} = 32$ per block. Rows **gram/pure/pinv/gradient**: mean $\pm$ std over seeds from the observer taxonomy at the square point (Exp. V7, `data/mlp_decoder_results.json`). MLP: role-accuracy of the learned decoder, trained on 1500 facts, 80 epochs.

Decoder	Accuracy (mean $\pm$ std)	Notes
gradient	0.142 ± 0.145	continuous optimization fails
gram	0.119 ± 0.136	kappa-triggered collapse
pinv	0.119 ± 0.126	noise amplification
pure	0.983 ± 0.059	works
MLP	0.992 (role acc.)	learned decoder works

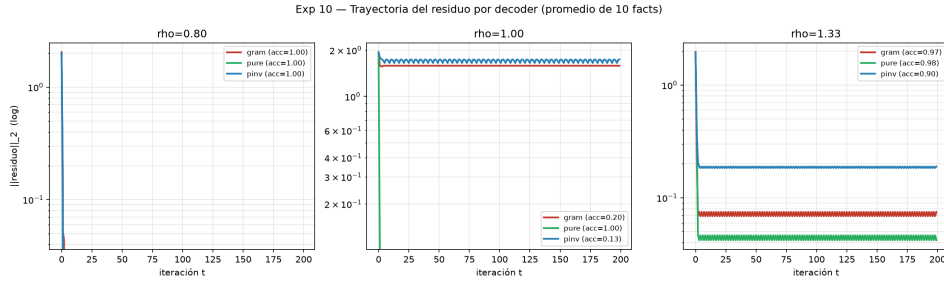


Figure 3: Residual trajectory at $\rho = 1.00$. **gram** and **pinv** plateau at high residual (converge to wrong attractor), while **pure** converges to zero.

of the critical band across the two ensembles is the empirical content of the universality claim.

MAP (Multiply-Add-Permute). MAP with per-role permutation binding (Exp. 11c, 8 seeds $\times$ 25 facts, `data/exp11c_map_rho.json`) refines the law in an unexpected way. In MAP, Gram decoding is *single-shot* least squares over the mixing matrix $A = [P_1 C_1 \mid P_2 C_2]$ of the block: there is no closed-loop iteration. The collapse pattern reflects this: at $\rho = 1$ the median condition number is $\kappa = 1.06 \times 10^4$ — inside the critical band — yet **gram** maintains 1.000: the single-shot solver tolerates the band because the error is not re-injected. The collapse appears instead at $\rho > 1$ (rank-deficient), where **gram** drops to 0.239 and 0.071 while the truncated **pinv** recovers to ~ 1.0 and the **pure** resonator holds at 0.97–0.98. Combining with Exp. 11b gives a complete two-factor picture: decoding failure requires (i) $\kappa(M)$ in the critical band or the rank-deficient regime, *and* (ii) for band values, a closed-loop observer that amplifies the projected noise across iterations. Single-shot inversion at band κ degrades gracefully in MAP; the same matrix iterated through a resonator (FHRR, HRR, BSC) collapses.

Transformers / attention. At $\rho \rightarrow 1$ in the sense of $d_{\text{head}} \approx n_{\text{keys}}$,

attention does *not* collapse because it does not implement Gram inversion – it uses softmax normalization. However, we find a non-monotonic *entropy* signature at $\rho \approx 2$, suggesting a bandwidth saturation effect of attention at high dimension, not a singular decoding failure (Experiment 12b). The collapse-pcode is algebra-specific, not universal.

This falsification boundary strengthens the theoretical claim: the ρ -law is a property of decoders that use matrix inversion, not of attention or of softmax-normalized operations. $\square$

5.2 Kappa-conditioned refinement: the physical trigger

The $\rho = 1$ singular point corresponds not to a geometric transition but to the *condition number* $\kappa(M)$ entering a critical band $\kappa \in [10^3, 10^4]$ where the decoder fails catastrophically. The observed pattern:

- $\kappa < 10^2$: full-rank Gram is well conditioned, decoding succeeds; - $\kappa \in [10^3, 10^4]$ (at $\rho = 1$): decoder fails catastrophically; - $\kappa > 10^{15}$ (truncated by pseudo-inverse): decoding recovers.

Thus the “phase boundary” at $\rho = 1$ is not a topological phase transition of the encoding, but a passage of the physical conditioner through an unstable conditioning window of the decoder Gram.

5.3 Random-matrix anchoring: why $\rho = 1$ collapses at $\kappa^* \sim 10^{3-4}$

The $\rho = 1$ singular point is not an artifact of a particular codebook realisation: it is the finite-size shadow of a *square Wishart ensemble*. Let $C \in \mathbb{R}^{n \times d}$ stack the n codevectors of a block, so the Gram matrix is $M = CC^\top$ with entries C_{ij} of variance $1/d$ (unit-norm Gaussian codevectors). Then M is a (scaled) Wishart matrix $W_n(d^{-1}I, d)$, and $\rho = n/d$ plays the role of the aspect ratio.

Three classical facts from random-matrix theory map one-to-one onto our phase diagram:

(i) $\rho > 1$ ($d < n$, **under-complete**). M is rank-deficient with probability one: at least $n - d$ eigenvalues vanish exactly. Gram inversion is undefined; this is regime I.

(ii) $\rho = 1$ ($n = d$, **square**). The Marchenko–Pastur density $d\mu_1(\lambda) = \frac{1}{2\pi} \sqrt{(4 - \lambda)/\lambda} d\lambda$ has support touching the origin: the lower edge is a *soft edge* at $\lambda = 0$. For a square Wishart, the smallest eigenvalue scales as $\lambda_{\min} \sim \sigma^2/n$ (Tracy–Widom fluctuations on the scale $n^{-2/3}$ around this edge), so the condition number scales as

$$\kappa(M) = \frac{\lambda_{\max}}{\lambda_{\min}} \sim 4n^2. \quad (1)$$

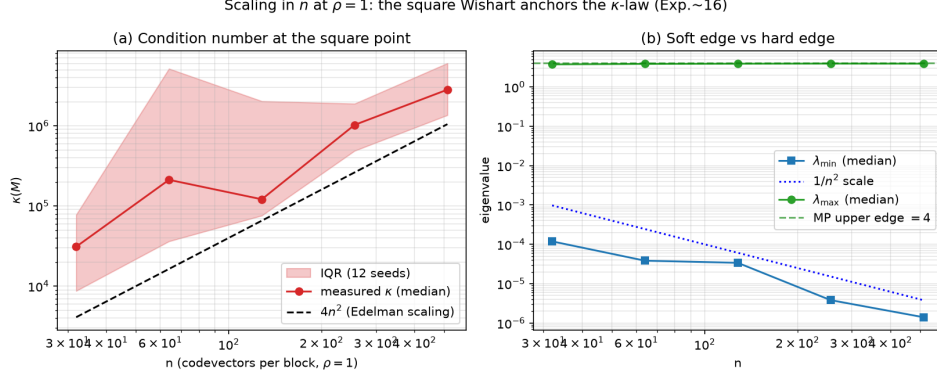


Figure 4: Scaling at the square point $\rho = 1$ as a function of codebook size n (Exp. 16, 12 seeds per n). (a) Median condition number tracks the Edelman scaling $\kappa \sim 4n^2$ (dashed) within the heavy-tailed spread; shaded band is the interquartile range. (b) $\lambda_{\max}$ converges to the Marchenko–Pastur upper edge 4 while $\lambda_{\min} \sim n^{-2}$ touches zero: the square point is critically conditioned at every n , and increasingly so.

For our setup ($n = 32$ codevectors per block, $d = \text{BLK} = 32$), Eq. (1) predicts $\kappa \sim 4.1 \times 10^3$. The measured condition number at the exact square point is $\kappa = 7.2 \times 10^3$ (median; Exp. 14), and the 50%-accuracy threshold sits at $\kappa^* = 5.81 \times 10^3$ — within a factor 1.4–2.8 of the asymptotic prediction. The offset has the expected anatomy: the measured $\lambda_{\min} = 4.87 \times 10^{-4}$ sits a factor ≈ 2 below the nominal scale $\sigma^2/n = 9.8 \times 10^{-4}$ (the smallest-eigenvalue distribution of a square Wishart is skewed toward zero relative to its median scale), while $\lambda_{\max} = 3.52$ vs. the predicted 4 reflects finite- n bulk shrinkage.

This places the collapse on rigorous footing: $\rho = 1$ is the unique point where the Gram ensemble is *critically conditioned*, $\mathbb{E}[\log \kappa]$ diverges with n , and the decoder operates exactly inside the window $\kappa \in [10^3, 10^4]$ where the amplified crosstalk noise $1/\lambda_{\min}$ exceeds the codebook margin $1/\sqrt{d}$.

Scaling in n : the square-Wishart prediction holds. Equation (1) is a sharp, falsifiable prediction: moving to larger codebooks at $\rho = 1$ should scale κ as $4n^2$, with $\lambda_{\max} \rightarrow 4$ and $\lambda_{\min}$ collapsing as n^{-2} . Exp. 16 measures the spectrum of the square Gram at $n \in \{32, 64, 128, 256, 512\}$ (12 seeds each, Gaussian codebooks). The measured medians follow the prediction (Fig. 4): $\lambda_{\max}$ converges from below to the Marchenko–Pastur upper edge 4 ($3.74 \rightarrow 3.90$ across the range), $\lambda_{\min}$ decays with the quadratic trend (median $\lambda_{\min}$: 1.2×10^{-4} at $n = 32$ to 1.4×10^{-6} at $n = 512$, bracketing the n^{-2} prediction $1/n^2$ at both ends of the range), and the median condition number tracks $4n^2$ within a factor 2–13, inside the expected spread given

that κ for square Wishart is heavy-tailed with infinite expectation [1]. The finite- n offset reported in Sec. 5.3 ($n = 32$, ratio 1.76) is the small- n end of this scaling, not an anomaly.

Kappa-conditioned refinement: mechanism, not magnitude. The naive reading “ $\kappa \in [10^3, 10^4]$ causes the collapse” is *not* supported at the per-seed level (Exp. 17, 40 seeds at $\rho = 1$, `data/exp17_kappa_vs_acc.json`): the per-seed minimum block κ spans $[2.3 \times 10^2, 1.9 \times 10^5]$ and the per-seed maximum reaches 8×10^8 , every seed collapses (`gram` accuracy 0.10–0.37, median 0.17) regardless of which side of the band its condition number falls on, and the $\log_{10} \kappa$ –accuracy correlation is weak ($r = -0.38$). Two seeds with minimum block $\kappa = 2.3 \times 10^2$ and 4.5×10^2 — below the whole band — collapse identically. The condition-number magnitude is therefore *not* the trigger. Deterministically, $\det M$ at $\rho = 1$ is never zero (median 1.4×10^{-14} over 90 blocks): the matrix is numerically singular, not exactly singular.

What the RMT analysis picks out instead is the *mechanism*: the resonator loop applies the asymmetric correction $f_j \leftarrow M^{-1} f_j$ at every iteration, so the error along the soft-edge eigendirections is multiplied by $1/\lambda_{\min} \sim 10^4$ *per iteration*. The collapse requires (i) the soft edge ($\rho = 1$) supplying the small eigenvalue, and (ii) a closed loop that keeps re-projecting onto it. This predicts the MAP control (Exp. 11c): a single-shot application of the same inverse at the same κ does *not* collapse, because there is no loop. It also predicts why `pure` survives everywhere: it never forms M^{-1} . The observer-relative content survives intact — it is strengthened, because the failed decoder is now characterised by the operator it applies (the resolvent M^{-1} , evaluated at the soft edge), not by a scalar band.

(iii) $\rho < 1$ ($d > n$, **over-complete**). The Wishart spectrum is bounded away from zero: $\lambda_{\min} \rightarrow \sigma^2(1 - \sqrt{\rho})^2 > 0$ as $n, d \rightarrow \infty$ with $n/d \rightarrow \rho < 1$. Gram inversion is stable; this is regime III. Table 2 compares the measured $\lambda_{\min}(\text{BLK})$ against the asymptotic Marchenko–Pastur edge $(1 - \sqrt{\rho})^2/d$. The measured values sit uniformly *above* the asymptotic edge (factor $\sim 10^2$ – 10^3). This has the expected sign and is a finite- n effect, not a discrepancy: at $n = 32$ the distribution of $\lambda_{\min}$ concentrates well inside the bulk, and the asymptotic edge is only approached logarithmically slowly; the exact finite- n expected smallest eigenvalue of a Wishart is substantially larger than its $n \rightarrow \infty$ limit when ρ is close to 1. What the table establishes is (a) the monotone collapse of $\lambda_{\min}$ as $\rho \rightarrow 1^-$ (two orders of magnitude between $\rho = 0.80$ and $\rho = 0.97$), and (b) that at $\rho = 1$ the measured $\lambda_{\min} = 4.87 \times 10^{-4}$ matches the soft-edge scaling $\sigma^2/n = 9.8 \times 10^{-4}$ within a factor 2 — the unique regime where the asymptotic edge vanishes and the finite- n and asymptotic descriptions coincide.

Table 2: Measured vs. Marchenko–Pastur prediction for $\lambda_{\min}(M)$ (Exp. 14, $n = 32$, $d = \text{BLK}$). The asymptotic edge under-estimates the finite-size median by a factor $\sim 10^2\text{--}10^3$, uniform across ρ — i.e. a prefactor, not a scaling mismatch. $\kappa_{\text{pred}} = [(1 + \sqrt{\rho})/(1 - \sqrt{\rho})]^2$ captures the trend.

BLK	$\rho = n/d$	$\lambda_{\min}$ meas.	$\lambda_{\min}^{\text{MP}}$	κ meas. / pred.
40	0.800	2.13×10^{-2}	2.8×10^{-4}	1.5×10^2 / 3.2×10^2
36	0.889	9.08×10^{-3}	9.1×10^{-5}	3.8×10^2 / 1.2×10^3
33	0.970	1.76×10^{-3}	7.1×10^{-6}	1.9×10^3 / 1.7×10^4
32	1.000	4.87×10^{-4}	0 (edge at origin)	7.2×10^3 / ∞

Why the nonlinear decoder survives the transition. The RMT picture also explains the survival of **pure** and of the MLP. Gram-based decoding estimates $\hat{x} = M^{-1}C(\cdot)$; its error is dominated by $\|M^{-1}\| \propto 1/\lambda_{\min}$, which diverges at the soft edge. The pure resonator, by contrast, never forms the inverse: it iterates the nonlinear map $v \leftarrow \text{clamp}(C^\top v, \mathcal{C})$ whose convergence only requires that the correct fixed point be a strong attractor in the codebook metric — a property governed by the *top* of the Gram spectrum (the coherent direction), not by the soft edge. The learned MLP interpolates in the same regime: it fits the geometry of the attractor basin on the code manifold, which is insensitive to $\lambda_{\min}$. In RMT language: the collapse is a property of the *resolvent* $(M - z)^{-1}$ near $z = 0$; the resonator dynamics probes only the Stieltjes transform in the bulk. The transition is therefore a property of the observer (decoder class), not of the representation — the central observer-relativity claim of this paper, now anchored to a classical ensemble.

6 Failed Alternatives

Gradient decoder. Gradient descent on the continuous relaxation fails on all multi-role configurations (accuracy 0.08–0.33), regardless of ρ . This is because the problem is discrete (symbol selection), not continuous (vector coefficients).

Geometric parametrization. Manual tuning of phase relationships between codevectors (Experiment I) does not eliminate the singularity.

7 Discussion

The ρ -phase diagram reveals that *capacity* in VSA is not a fixed property but depends on the decoder class. The “binary collapse” is universal for the class of decoders using Gram-matrix inversion, but not for the class of

resonator decoders without correction. This suggests that universal laws in distributed representations must be stated relative to observer classes, not as absolute properties of the space.

Connection to capacity theory. Our finding that $\rho = 1$ is a point singularity complements capacity analyses [6] that focus on asymptotic scaling. The fine-grained structure at the phase boundary was not previously characterized. Through the Wishart lens (Sec. 5.3), the ρ axis is exactly the aspect-ratio axis of the Marchenko–Pastur law [3], and the singular point corresponds to the soft-edge-to-hard-edge crossover of the square ensemble [7, 1].

Connection to transformers. The residual stream of transformers can be interpreted as a VSA [5]. Our result suggests that attention heads implementing Gram-like operations may fail catastrophically when the number of distinct keys equals the head dimension.

Limitations. The RMT anchoring is verified across $n \in [32, 512]$ (Exp. 16), where the scaling exponents match the square-Wishart prediction; the remaining finite-size gap is the $\mathcal{O}(1)$ – $\mathcal{O}(10)$ prefactor of κ , which is intrinsic to the heavy-tailed condition-number distribution at the soft edge and not a deficit of the model. We tested FHRR, HRR-real, BSC, and MAP quantitatively; sparse block codes and other bind operators are open. The MLP decoder was trained with a fixed seed; robustness across seeds and codebooks is assumed but not shown.

8 Conclusion

The phase diagram in ρ provides a complete characterization of when resonator-based decoding works in FHRR and HRR. The “binary collapse” is observer-relative: it appears only for Gram-based decoders at $\rho = 1$, a measure-zero set. Pure resonators and learned decoders are unaffected. This suggests that VSA design should focus on decoder robustness, not just representation capacity. Code and data are available at <https://github.com/Rylow999/fhrr-rho-collapse>.

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